# REAL LIFE EXCEL SHEETS

5 Complete Professional Sheets Used in Real Businesses Every Day
Full data + step-by-step instructions + every formula explained with WHY

| Sheet | Title | Used By | Key Formulas | Approx Time |
|---|---|---|---|---|
| 1 | Employee Salary Register | HR Dept · Every Company | IF, SUM, ROUND, VLOOKUP, MAX | 45 min |
| 2 | GST Sales Invoice Tracker | Accounts · GST-registered Biz | SUMIF, IF, ROUND, TODAY, VLOOKUP | 45 min |
| 3 | Loan EMI & Repayment Tracker | Banks · Finance · Home Buyers | PMT, IPMT, PPMT, CUMIPMT, IF | 30 min |
| 4 | Daily Cash Flow Tracker | Small Biz · Shop Owners · CA | SUM, SUMIF, IF, MAX, MIN, IFERROR | 30 min |
| 5 | Exam Result & Rank Sheet | Schools · Coaching · HR Tests | AVERAGE, RANK, COUNTIF, PERCENTILE, IF | 30 min |

**How to use this PDF:** Open Excel → Create a NEW sheet for each task → Type the column headers → Enter the sample data exactly as shown → Write each formula in the marked (✍) columns. Read the full explanation before writing any formula. Every step tells you WHAT to type AND WHY. After completing all 5 sheets, you will have a real working business toolkit in Excel.

# ■ Employee Salary Register

The payro

## ■ What Is This Sheet & Who Uses It?

The Employee Salary Register is the master document for monthly payroll processing. Every company — from a 5-person startup to a 50,000-employee corporation — needs this sheet. HR uses it to calculate take-home salary after all deductions. Accounts uses it to post the salary journal entry. The payslip handed to each employee is generated from this sheet. In real companies, this sheet is maintained by the HR Executive and checked by the Finance Manager before salaries are transferred on the 1st of each month.

### ■ Sheet Structure — Create These Columns

Sheet name: 'June 2025 Salary' | One row per employee

| Emp ID | Name | Dept | Basic ■ | HRA ■ | Conv ■ | Gross ■ | PF Ded ■ | ESI Ded ■ | PT ■ | TDS ■ | Total Ded ■ | Net Pay ■ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| E001 | Ravi Kumar | Sales | 20000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E002 | Priya Singh | Accounts | 12000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E003 | Aman Verma | Warehouse | 10000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E004 | Sunita Rao | Admin | 9000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E005 | Rohit Sharma | Sales | 15000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E006 | Kavya Patel | IT | 18000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E007 | Deepak Nair | Marketing | 13000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |
| E008 | Meena Joshi | Accounts | 11000 | ✍ | ✍ | ✍ | ✍ | ✍ | 200 | ✍ | ✍ | ✍ |

✍ = column where you write a formula. Fixed values: Professional Tax (PT) = ■200 for all with Gross > ■15,000

## ■ Step-by-Step: Every Formula Explained

**STEP 1 — HRA (House Rent Allowance) — Cell E2**

```
=ROUND(D2*0.4, 0)
```

HRA = 40% of Basic Salary. ROUND(..., 0) removes paise (decimal). Why 40%? HRA is typically 40% of Basic for non-metro and 50% for metro cities (Mumbai, Delhi, Bangalore, Chennai). Your company policy decides this. For Ravi (Basic ■20,000): HRA = 20,000 × 0.4 = ■8,000.

Copy this formula from E2 down to E9 for all employees. Each row automatically calculates 40% of that row's Basic.

**STEP 2 — Conveyance Allowance — Cell F2**

```
=IF(D2>=15000, 1600, 800)
```

Conveyance Allowance (travel to work) is given based on salary grade. Standard: ■1,600 for senior employees (Basic ≥ ■15,000) and ■800 for juniors. This is the IF formula in action — checking a condition and returning different values. The ■1,600 limit is tax-exempt under Income Tax Act (Section 10(14)).

Copy down to F9. Ravi (■20,000 Basic) → ■1,600. Priya (■12,000 Basic) → ■800.

**STEP 3 — Gross Salary — Cell G2**

```
=D2+E2+F2
```

Gross = Basic + HRA + Conveyance. This is the TOTAL earnings before any deduction. For Ravi: 20,000 + 8,000 + 1,600 = ■29,600. The Gross is used to check ESI eligibility (Gross ≤ ■21,000). Copy down to G9.

**STEP 4 — PF Deduction (Provident Fund) — Cell H2**

```
=ROUND(MIN(D2, 15000) * 0.12, 0)
```

Employee PF = 12% of Basic. BUT there is a wage ceiling — if Basic is above ■15,000, PF is capped at 12% of ■15,000 = ■1,800. The MIN(D2, 15000) picks the SMALLER of actual Basic and 15,000. For Ravi (Basic ■20,000): MIN(20000, 15000) = 15000 → PF = 15000 × 12% = ■1,800. For Priya (Basic ■12,000): MIN(12000, 15000) = 12000 → PF = 12000 × 12% = ■1,440.

**STEP 5 — ESI Deduction — Cell I2**

```
=IF(G2<=21000, ROUND(G2*0.0075, 0), 0)
```

ESI (Employee State Insurance) = 0.75% of Gross Salary. Applicable ONLY if Gross ≤ ■21,000. Above this limit, no ESI is deducted. The IF formula checks: Is Gross ≤ 21,000? If yes → calculate 0.75%. If no → 0 (zero). Ravi (Gross ■29,600) → ESI = 0. Priya (Gross ■13,040) → ESI = 13040 × 0.75% = ■98.

**STEP 6 — Professional Tax — Cell J2**

```
=IF(G2>15000, 200, IF(G2>10000, 150, 0))
```

Professional Tax is a STATE tax (Karnataka rates used here). Gross > ■15,000 → ■200/month. Gross ■10,001–■15,000 → ■150/month. Gross ≤ ■10,000 → ■0. Different states have different slabs. Since we fixed ■200 in the data, you can also just type 200 directly — but the formula is the right professional approach.

**STEP 7 — TDS (Tax Deducted at Source) — Cell K2**

```
=IFERROR(MAX(0, ROUND((G2*12 - 250000) * 0.05 / 12, 0)), 0)
```

TDS is simplified here for learning. Annual Gross = Monthly Gross × 12. Standard Deduction = ■50,000 (approximate). Taxable Income = Annual Gross – ■2,50,000 (basic exemption). Tax = 5% on ■2.5L–■5L slab. Monthly TDS = Annual Tax ÷ 12. MAX(0,...) ensures no negative TDS. IFERROR handles any calculation errors. For real payroll, consult a CA for proper TDS calculation with Form 12BB.

**STEP 8 — Total Deductions — Cell L2**

```
=H2+I2+J2+K2
```

Sum of all deductions: PF + ESI + Professional Tax + TDS. This total is subtracted from Gross to get take-home pay.

**STEP 9 — Net Pay (Take-Home Salary) — Cell M2**

```
=G2-L2
```

Net Pay = Gross – Total Deductions. This is the actual amount transferred to the employee's bank account on salary day. For Ravi: ■29,600 – (■1,800 + ■0 + ■200 + ■TDS) = actual take-home.

**STEP 10 — Summary Row at the Bottom (Row 11)**

```
=SUM(D2:D9) =MAX(M2:M9) =MIN(M2:M9) =AVERAGE(M2:M9)
```

Add a TOTALS row: SUM each salary column to get total payroll cost. Add MAX to see highest net pay, MIN for lowest, AVERAGE for average salary. These summary figures are used for the Salary Journal Entry in Tally and for management reporting.

■ *REAL WORLD: This exact sheet is maintained by every HR department. Before transferring salaries, the Finance Manager cross-checks: Total Net Pay column = total bank transfer amount. Any mismatch means there's an error somewhere. In large companies, this sheet feeds into the payroll software automatically. Your Excel version IS the payroll software for small businesses!*

■■ *LEGAL NOTE: PF must be deposited by 15th of next month. ESI by 15th. TDS by 7th. Professional Tax by end of month. Missing these deadlines attracts penalties and interest. Set reminders!*

# ■ GST Sales Invoice Tracker

The regist

## ■ What Is This Sheet & Who Uses It?

Every business registered under GST (Goods & Services Tax) must maintain a Sales Register showing invoice-wise details of all sales made. This data is filed with the government every month as GSTR-1 (by the 11th of next month). The Tax Department uses this to cross-check with your customers' purchase records. An error in this sheet = mismatch in GST returns = tax notice. Accountants, shop owners, exporters, and every trading business maintain this daily. It also forms the basis for calculating Output GST payable to the government.

## ■ Sheet Structure — Create These Columns

| ate | Customer | GSTIN | State | Type | Item | Qty | Rate ■ | Taxable ■ | GST% | CGST ■ | SGST ■ | IGST ■ | Invoice Total ■ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -Jun | Walk-in Cash | - | KA | B2C Cash | Mobile Cover | 10 | 200 | ✍ | 18 | ✍ | ✍ | - | ✍ |
| -Jun | Priya Stores | 29ABCDE1234Z1 | KA | B2B Intra | Phone Stand | 25 | 150 | ✍ | 18 | ✍ | ✍ | - | ✍ |
| -Jun | Mumbai Electronics | 27FGHIJ5678K1 | MH | B2B Inter | USB Cable | 50 | 140 | ✍ | 18 | - | - | ✍ | ✍ |
| -Jun | Local Retail | 29KLMNO9012Z1 | KA | B2B Intra | Charger | 15 | 300 | ✍ | 18 | ✍ | ✍ | - | ✍ |
| -Jun | Delhi Distributor | 07PQRST3456Z1 | DL | B2B Inter | Mobile Cover | 40 | 200 | ✍ | 18 | - | - | ✍ | ✍ |
| -Jun | Export Customer | - | US | Export | USB Cable | 100 | 140 | ✍ | 0 | - | - | - | ✍ |
| -Jun | Ahmedabad Trader | 24UVWXY7890Z1 | GJ | B2B Inter | Screen Guard | 30 | 120 | ✍ | 18 | - | - | ✍ | ✍ |

## ■ Step-by-Step: Every Formula Explained

**STEP 1 — Taxable Amount — Column J**

```
=H2*I2
```

Taxable Amount = Quantity × Rate per unit. This is the SALE VALUE before adding GST. SI-001: 10 covers × ■200 = ■2,000. This is the base on which GST is calculated. Always separate the taxable value from GST on your invoice — it is a legal requirement under GST rules.

**STEP 2 — CGST (Central GST) — Column L**

```
=IF(E2="KA", ROUND(J2*K2/2/100, 2), 0)
```

CGST applies ONLY for Intra-State sales (both buyer and seller in same state). Karnataka = 'KA'. If state = KA, CGST = Taxable × (GST Rate / 2). Why divide by 2? Because for intra-state, total GST = CGST + SGST, and both are equal halves. SI-001 (KA): 2000 × 18/2/100 = ■180. SI-003 (MH, inter-state): CGST = 0.

**STEP 3 — SGST (State GST) — Column M**

```
=IF(E2="KA", ROUND(J2*K2/2/100, 2), 0)
```

SGST is always equal to CGST for the same transaction. Use the EXACT same formula as CGST. SGST and CGST together = total GST for intra-state. For SI-002 (KA, ■3,750 taxable at 18%): CGST = ■337.50, SGST = ■337.50, Total GST = ■675.

**STEP 4 — IGST (Integrated GST) — Column N**

```
=IF(E2<>"KA", ROUND(J2*K2/100, 2), 0)
```

IGST applies for Inter-State sales (buyer in a different state). Full GST rate applied as IGST. If state ≠ KA, IGST = Taxable × GST Rate. SI-003 (MH): IGST = 7,000 × 18% = ■1,260. For exports (GST Rate = 0), IGST = 0. IGST goes entirely to Central Government initially.

**STEP 5 — Invoice Total — Column O**

```
=J2+L2+M2+N2
```

Invoice Total = Taxable + CGST + SGST + IGST. Since intra-state has no IGST and inter-state has no CGST/SGST, this formula works perfectly for both — because the unused columns will be 0. The Invoice Total is the amount the customer pays you.

**STEP 6 — Summary Table (below your data, Row 11+)**

```
=SUMIF(F2:F8, "B2B Intra", O2:O8) =SUMIF(F2:F8, "B2B Inter", O2:O8)
```

```
=SUM(L2:L8) =SUM(M2:M8) =SUM(N2:N8)
```

Create a summary section: B2C Sales Total, B2B Intra-State Total, B2B Inter-State Total, Export Total. This summary IS your GSTR-1 breakup. Also total CGST, SGST, and IGST collected — these are your Output Tax liabilities. SUMIF with the Type column lets you slice by sale type.

**STEP 7 — Overdue Invoice Alert — Column Q (add this!)**

```
=IF(P2="Credit", IF(TODAY()-B2>45, "■ Overdue", IF(TODAY()-B2>30, "■ Due Soon", "■ OK")), "-")
```

This checks: Is it a Credit sale? If yes, how many days since invoice? Over 45 days → Red Overdue alert. 30–45 days → Yellow due soon. Under 30 days → Green OK. This turns your sales register into an accounts receivable tracker too. One sheet does two jobs!

■ *REAL WORLD: GSTR-1 (due 11th of every month) requires EXACTLY this data — Invoice number, date, customer GSTIN, state, taxable value, CGST/SGST/IGST separately. Your Excel register IS the GSTR-1 raw data. In Tally, F8 Sales vouchers auto-populate this register — now you know every field Tally is tracking!*

# ■ Loan EMI & Repayment Tracker



## ■ What Is This Sheet & Who Uses It?

Anyone who has taken a loan — home loan, car loan, business loan, personal loan — needs an EMI tracker. Banks give you a loan statement but it doesn't tell you: How much total interest have I paid so far? How much principal is still outstanding? If I pay an extra ■5,000 this month, how much will I save? This Excel sheet answers all these questions. It builds a complete Amortisation Schedule — the table that shows for EVERY month: how much goes to interest, how much reduces principal, and what balance remains. Finance teams use this for business loans. Homebuyers use it to plan prepayment. Banks prepare this table for every loan.

### ■ Part A: Loan Details Setup

| Parameter | Value | Notes |
|---|---|---|
| Loan Amount (■) | 2000000 | ■20 Lakhs home loan |
| Annual Interest Rate | 8.50% | Bank lending rate |
| Loan Period (Years) | 20 | 240 months total |
| Monthly Interest Rate (✍) | =B2/12 | 8.5%/12 = 0.708% per month |
| Total Months (✍) | =B3*12 | = 240 months |
| Monthly EMI (✍) | =PMT(B4,B5,-B1) | Use PMT formula — answer: ■17,398 |
| Total Amount Payable (✍) | =B6*B5 | EMI × Total Months |
| Total Interest Cost (✍) | =B7-B1 | Total Paid – Principal |

### ■ Part B: Amortisation Schedule (Build rows for first 12 months minimum)

| Month | Opening Balance ■ | EMI ■ | Interest ■ ✍ | Principal ■ ✍ | Closing Balance ■ | Cumulative Interest ■ | Cumulative Principal ■ |
|---|---|---|---|---|---|---|---|
| 1 | =B1 | =B6 | ✍ | ✍ | ✍ | ✍ | ✍ |
| 2 | =prev closing | =B6 | ✍ | ✍ | ✍ | ✍ | ✍ |
| 3 | ... | =B6 | ... | ... | ... | ... | ... |
| ... | ... | =B6 | ... | ... | ... | ... | ... |
| 240 | ... | =B6 | ... | ... | =0 (loan closed) | ... | ... |

## ■ Step-by-Step: Every Formula Explained

**STEP 1 — Monthly EMI using PMT — Cell B6**

```
=PMT(B4, B5, -B1)
```

PMT = Payment. This calculates the fixed monthly payment (EMI) for a loan. B4 = monthly interest rate (8.5%/12 = 0.00708). B5 = total number of months (240). B1 = loan amount (■20,00,000). The negative sign (–B1) is because you RECEIVED the money (inflow). PMT returns a positive number = what you PAY each month. Answer: ■17,398. This is an exact mathematical calculation — same as what banks use!

**STEP 2 — Interest Portion of each EMI — Column D**

```
=ROUND(B_OpeningBalance * $B$4, 0)
```

Interest for each month = Opening Balance × Monthly Interest Rate. For Month 1: 20,00,000 × 0.00708 = ■14,167. This is why early EMIs are mostly interest — you're paying interest on the FULL loan amount. By Month 240, interest is tiny (< ■100) because the outstanding balance is almost zero.

**STEP 3 — Principal Portion of each EMI — Column E**

```
=EMI - Interest_Amount (i.e., =$B$6 - D_row)
```

Principal = EMI – Interest. Every rupee of EMI that is NOT interest reduces your loan balance. Month 1: ■17,398 – ■14,167 = ■3,231 principal. Month 240: ■17,398 – ■122 = ■17,276 principal. As the loan ages, principal portion GROWS and interest portion SHRINKS.

**STEP 4 — Closing Balance — Column F**

```
=Opening_Balance - Principal_Amount
```

Closing Balance = Opening – Principal paid this month. This becomes next month's Opening Balance. Month 1: 20,00,000 – 3,231 = ■19,96,769. After 240 months, this should be exactly ■0. If not, there's a rounding error — use ROUND in your formulas.

**STEP 5 — Cumulative Interest & Principal — Columns G and H**

```
=G_prev + D_current =H_prev + E_current
```

Running totals of interest paid and principal repaid since Day 1. After 5 years (60 months): check how much interest you've already paid vs principal. Most people are shocked — by year 5, you've paid ~60% of your total interest but only reduced principal by ~10%! This insight motivates prepayment.

**STEP 6 — Add Prepayment Analysis Row**

```
=NPER($B$4, -$B$6, Outstanding_Balance_After_Prepayment)
```

After any month, if you make a lump-sum prepayment, enter it in a special cell. Then use NPER to calculate: how many months are now remaining? Compare with original remaining months → months saved = interest saved. E.g., after 5 years, prepay ■2,00,000. NPER shows you finish in 170 months instead of 180 — save 10 months × ■17,398 = ■1,73,980 interest saved!

■ *REAL WORLD: Every bank's 'Loan Statement' that comes annually IS this amortisation schedule. Now you can build it yourself and understand EXACTLY where your money goes. Many homebuyers pay EMI for 20 years and realise they paid more interest than the actual loan! This sheet opens your eyes.*

# ■ Daily Cash Flow Tracker

For shops

## ■ What Is This Sheet & Who Uses It?

The Daily Cash Flow Tracker is the most fundamental financial tool for any small business. A shopkeeper, a freelancer, a restaurant owner, a medical clinic — everyone needs to know: How much cash came in today? How much went out? What is today's closing balance? Is my cash balance increasing or decreasing over the month? This sheet is also known as a 'Cash Book' — the most basic accounting record. Auditors, tax officers, and bank managers always ask for the Cash Book first. CA firms maintain this for clients. Shop owners maintain it daily. This is the same sheet as the Day 4 Cash Book but now made much more detailed with categories and analysis.

### ■ Part A: Daily Transaction Log

| Date | Particulars | Category | Mode | Receipt ■ | Payment ■ | Balance ■ | Day Net ■ | Notes |
|---|---|---|---|---|---|---|---|---|
| 01-Jun | Opening Balance | - | - | 50000 | - | ✍ | - | Month opening |
| 01-Jun | Sales — Cash | Sales Revenue | Cash | 2360 | - | ✍ | ✍ | - |
| 01-Jun | Rent Paid | Rent | Cash | - | 8000 | ✍ | ✍ | - |
| 02-Jun | Sales — UPI | Sales Revenue | UPI | 4425 | - | ✍ | ✍ | - |
| 02-Jun | Stock Purchase | Purchases | Cash | - | 4720 | ✍ | ✍ | - |
| 03-Jun | Salary Paid | Salaries | Bank Transfer | - | 30900 | ✍ | ✍ | - |
| 05-Jun | Sales — Cash | Sales Revenue | Cash | 3500 | - | ✍ | ✍ | - |
| 05-Jun | Electricity Bill | Utilities | Cash | - | 2400 | ✍ | ✍ | - |
| 07-Jun | Customer Receipt | Accounts Receivable | Bank | 22000 | - | ✍ | ✍ | Ravi Kumar payment |
| 10-Jun | Marketing Ads | Marketing | Online | - | 5000 | ✍ | ✍ | Facebook Ads |
| 15-Jun | Sales — Cash | Sales Revenue | Cash | 6200 | - | ✍ | ✍ | - |
| 20-Jun | Supplier Payment | Creditor Payment | Bank | - | 18000 | ✍ | ✍ | Sharma Traders |

## ■ Step-by-Step: Every Formula Explained

**STEP 1 — Opening Balance in G2 (very first row)**

```
=E2
```

For the very first row (Opening Balance), the Balance column simply equals the Receipt amount (■50,000). This sets the starting point for all subsequent calculations.

**STEP 2 — Running Balance — Column G (all rows after row 2)**

```
=G_previous + E_current - F_current
```

Balance = Previous Balance + Today's Receipt – Today's Payment. This is the RUNNING BALANCE — after every transaction, what is the current cash/bank position? G3 = G2 + E3 – F3 = 50,000 + 2,360 – 0 = ■52,360. G4 = G3 + E4 – F4 = 52,360 + 0 – 8,000 = ■44,360. This formula captures the cascading nature of cash flow.

**STEP 3 — Day Net Flow — Column H**

```
=IFERROR(E_current - F_current, 0)
```

Day Net = Receipt – Payment for that specific row. Positive = net cash in, Negative = net cash out. Useful for seeing which days were good cash flow days and which were bad. IFERROR prevents errors when cells are empty.

**STEP 4 — Monthly Summary Table (create below the daily log)**

```
=SUMIF(C:C, "Sales Revenue", E:E) [Total Sales Receipts]
```

```
=SUMIF(C:C, "Purchases", F:F) [Total Purchases Paid]
```

```
=SUMIF(C:C, "Rent", F:F) [Total Rent]
```

```
=SUMIF(C:C, "Salaries", F:F) [Total Salaries]
```

SUMIF is the hero of this analysis. It sums only the rows where Category matches. Create one SUMIF per category. Result: a beautiful Income & Expenditure statement broken by category — automatically built from your daily transactions. This IS your monthly P&L; summary!

**STEP 5 — Cash Position Alert**

```
=IF(G_last<10000, "■ LOW CASH – Replenish!", IF(G_last<25000, "■■ Watch Balance", "■ Healthy"))
```

Place this formula at the top of your sheet (cell A1 or a prominent location). It checks the LAST balance row and flags if cash is dangerously low. Below ■10,000 → Red alert. Below ■25,000 → Yellow watch. Above ■25,000 → Green. This is your live business health indicator — check it every morning!

**STEP 6 — Mode-wise Receipt Summary**

```
=SUMIF(D:D, "Cash", E:E) =SUMIF(D:D, "UPI", E:E) =SUMIF(D:D, "Bank", E:E)
```

How much came in as Cash vs UPI vs Bank Transfer? This analysis helps you understand your payment mix. High Cash → needs more frequent banking. High UPI → digital business (no cash handling risk). This is important for GST compliance — cash transactions above ■2 lakh require PAN of customer.

■ *REAL WORLD: This sheet works as: (1) Daily Cash Book for accounting, (2) Monthly P&L; summary via SUMIF, (3) Cash position tracker, (4) Bank reconciliation source. The Income Tax Department asks for bank statements + cash book during audit. This sheet IS your cash book. A CA reviewing your accounts will check this first!*

# ■ Exam Result & Rank Sheet



## ■ What Is This Sheet & Who Uses It?

Every school, college, coaching centre, and HR department that conducts exams needs a Result Sheet. Teachers spend hours manually calculating averages, ranks, and grades. This sheet automates everything. Enter marks once → all results, ranks, grades, pass/fail, top performers, weak students — all auto-calculated. School teachers use this every semester. Coaching institutes use it after every mock test. Corporate HR teams use it after aptitude tests for hiring. This is one of the most commonly built Excel sheets — and it showcases RANK, AVERAGE, COUNTIF, PERCENTILE, and VLOOKUP working together.

### ■ Part A: Student Marks Sheet

| Roll No | Name | Maths /100 | Science /100 | English /100 | Hindi /100 | Social /100 | Total /500 | Average ✍ | Grade ✍ | Rank ✍ | Pass/Fail ✍ | Percentile ✍ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 101 | Aarav Mehta | 85 | 78 | 92 | 88 | 76 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 102 | Priya Sharma | 92 | 95 | 88 | 91 | 94 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 103 | Rajan Verma | 45 | 52 | 48 | 55 | 60 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 104 | Sunita Rao | 78 | 82 | 75 | 80 | 85 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 105 | Karan Patel | 65 | 70 | 68 | 72 | 65 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 106 | Divya Nair | 96 | 92 | 98 | 94 | 97 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 107 | Rohit Singh | 55 | 48 | 60 | 52 | 58 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 108 | Meena Joshi | 88 | 85 | 90 | 82 | 79 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 109 | Arjun Kumar | 72 | 68 | 75 | 70 | 74 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |
| 110 | Nisha Gupta | 40 | 38 | 45 | 42 | 50 | ✍ | ✍ | ✍ | ✍ | ✍ | ✍ |

## ■ Step-by-Step: Every Formula Explained

**STEP 1 — Total Marks — Column H**

```
=SUM(C2:G2)
```

Sum all 5 subject marks. This is Aarav's Total out of 500. Copy down for all 10 students. Simple SUM — but the foundation for all other calculations.

**STEP 2 — Average Marks — Column I**

```
=ROUND(H2/5, 1) OR =ROUND(AVERAGE(C2:G2), 1)
```

Average = Total ÷ 5 subjects. ROUND to 1 decimal place. Aarav: 419/5 = 83.8. This average is what goes on the report card and is used for grade calculation.

**STEP 3 — Grade — Column J**

```
=IF(I2>=90,"A+", IF(I2>=80,"A", IF(I2>=70,"B+", IF(I2>=60,"B", IF(I2>=50,"C", "F")))))
```

This nested IF checks average marks and assigns grade. 90+ = A+, 80–89 = A, 70–79 = B+, 60–69 = B, 50–59 = C, below 50 = F (Fail). 5 levels of nesting — one IF inside another. You can also do this with a Grade Table + VLOOKUP (shown below as an alternative).

**ALTERNATIVE GRADE METHOD using VLOOKUP + Grade Table**

Create a Grade Table in a separate area:

| Min Score | Grade |
|---|---|
| 0 | F |
| 50 | C |
| 60 | B |
| 70 | B+ |
| 80 | A |

| | |
|---|---|
| 90 | A+ |

```
=VLOOKUP(I2, $P$2:$Q$7, 2, 1)
```

The '1' at end means approximate match — VLOOKUP finds the highest value ≤ Average. Much cleaner than nested IF when you have many grade levels! The $ signs lock the Grade Table range.

**STEP 4 — Rank — Column K**

```
=RANK(H2, $H$2:$H$11, 0)
```

RANK(value, range, 0) gives position among all students. 0 = rank from highest (Rank 1 = topper). The $ signs lock the total range so when you copy down, the range stays fixed but H2 changes to H3, H4, etc. Divya (highest total) → Rank 1. Nisha (lowest) → Rank 10.

**STEP 5 — Pass / Fail — Column L**

```
=IF(AND(C2>=33, D2>=33, E2>=33, F2>=33, G2>=33, H2>=165), "■ PASS", "■ FAIL")
```

Pass condition: Minimum 33% in EACH subject AND minimum 33% total (165/500). The AND function checks ALL conditions — if even ONE fails, the student fails. Rajan has 45 in Maths (passes) but check all subjects. Nisha has 40 in Maths — barely passes that subject but overall average is low.

You can also add: Fail in how many subjects? =COUNTIF(C2:G2, '<33') gives the count of subjects failed.

**STEP 6 — Percentile Rank — Column M**

```
=ROUND(PERCENTRANK($H$2:$H$11, H2)*100, 1)
```

Percentile tells you: what % of students scored BELOW this student. Divya (top scorer) → 100th percentile (everyone else is below her). Nisha (bottom scorer) → 0th percentile. Middle student → ~50th percentile. This is used in competitive exams (JEE, NEET, CAT) where absolute marks don't matter — your RELATIVE position does!

**STEP 7 — Class Analysis Summary (create below the marks table)**

```
=MAX(H2:H11) [Highest Total] =MIN(H2:H11) [Lowest Total]
=AVERAGE(H2:H11) [Class Average] =MEDIAN(H2:H11) [Median Score]
=COUNTIF(L2:L11, "■ PASS") [Number of Students Passed]
=COUNTIF(L2:L11, "■ FAIL") / COUNTA(L2:L11) * 100 [Failure Rate %]
=PERCENTILE(H2:H11, 0.9) [90th Percentile Score — entry bar for top tier]
```

These 8 formulas give you the COMPLETE class analysis in one block. Teachers present this in parent meetings. Coaches use it to identify which students need extra coaching. HR teams use pass rate to calibrate test difficulty.

**STEP 8 — Subject-wise Analysis (for teachers)**

```
=AVERAGE(C2:C11) [Maths class average] =MIN(C2:C11) [Lowest Maths score]
=COUNTIF(C2:C11, "<33") [Students failed in Maths]
```

Do this for all 5 subjects. If Maths average is 68 but Hindi average is 85, the Maths teacher needs to change teaching methods. Subject-wise analysis identifies curriculum gaps and teacher effectiveness.

■ *REAL WORLD: This exact sheet is used by every school in India. CBSE board exam results use PERCENTILE (not rank) to compare performance across different question sets. JEE ranks 13 lakh students — all using RANK and PERCENTILE formulas at massive scale. Your Excel sheet is a miniature version of that system!*

## ■ WHAT YOU'VE BUILT — ALL 5 SHEETS SUMMARY

| Sheet | Real-World Purpose | Key Insight You Gained |
|---|---|---|
| 1 — Salary Register | Monthly payroll for any company | PF cap ■15K, ESI eligibility ■21K, TDS annual projection |
| 2 — GST Invoice Tracker | GSTR-1 raw data, daily sales book | CGST+SGST for same state, IGST for different state, 0% for exports |

| | | |
|---|---|---|
| 3 — Loan EMI Tracker | Home loan, car loan, business loan | Early EMIs = mostly interest. PMT+IPMT+PPMT work together |
| 4 — Cash Flow Tracker | Daily cash book, monthly P&L; | SUMIF by category = automatic income-expense statement |
| 5 — Exam Result Sheet | School results, HR aptitude test | RANK+PERCENTILE = relative position. AND for multi-condition pass |

**Excel is not just software — it is how business runs. Master these 5 sheets and you master real business accounting.**